Homework Assignment 3: Oyster Restoration in EEZ

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```
library(tidyverse)
library(tmap)
library(sf)
library(terra)
library(knitr)
library(kableExtra)

sst_2008 <- rast("data/average_annual_sst_2008.tif")
sst_2009 <- rast("data/average_annual_sst_2009.tif")
sst_2010 <- rast("data/average_annual_sst_2010.tif")
sst_2011 <- rast("data/average_annual_sst_2011.tif")
sst_2012 <- rast("data/average_annual_sst_2012.tif")

depth <- rast("data/depth.tif")

eez <- st_read("data/wc_regions_clean.shp")
```

```
Reading layer `wc_regions_clean' from data source
  `C:\Users\aaron\Documents\Git\oyster-restoration-eds223\data\wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 0 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
CRS:           NA
```

Check coordinate reference system

```
if (crs(sst_2008) == crs(depth)) {
    print('crs match')
} else {
    warning('crs does not match')
}
```

```
[1] "crs match"
```

```
if (crs(sst_2008) == crs(eez)) {
    print('crs match')
} else {
    warning('crs does not match')
}
```

```
# match eez dataframe to sst raster
eez <- st_set_crs(eez, 4326)
```

Process data:

Next, we need to process the SST and depth data so that they can be combined. In this case the SST and depth data have slightly different resolutions, extents, and positions.

- find the mean SST from 2008-2012 (e.g. create single raster of average SST)
- convert average SST from Kelvin to Celsius
  - hint: subtract by 273.15
- crop depth raster to match the extent of the SST raster
- note: the resolutions of the SST and depth data do not match
  - resample the depth data to match the resolution of the SST data using the bilinear method approach
- check that the depth and SST match in resolution, extent, and coordinate reference system
  - hint: can the rasters be stacked?

```r
# stack the sst rasters and obtain the average sst from 2008-2012
# do not use merge() because it combines rasters spatially joining by the geographic covers
sst <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
sst_avg_K <- mean(sst, na.rm = TRUE)

# sst_avg_K <- global(sst, 'mean', na.rm = TRUE)
## cannot use global() because it converts the raster into a dataframe

# convert sst unit from Kelvin to Celsius
sst_avg_C <- sst_avg_K - 273.15
```

Match extent

```r
# crop depth raster to match the extent of the sst raster
depth_cropped <- crop(depth, sst_avg_C)
```

Match resolution

```r
# resample depth raster to match the sst raster's resolution
depth_cropped <- resample(x = depth_cropped, y = sst_avg_C, method = 'bilinear')
```

Stack depth and sst

```r
# depth and SST have matching resolution, extent, and coordinate reference system
combined_rast <- c(depth_cropped, sst_avg_C)
```

Find suitable locations:

To find suitable locations for marine aquaculture, we'll need to find locations that are suitable in terms of both SST and depth.

- reclassify SST and depth data into locations that are suitable for oysters
  - hint: set suitable values to 1 and unsuitable values to 0
- find locations that satisfy both SST and depth conditions

Table 1: Environmental Criteria for Oyster Suitability

Variable	Suitable Range
Sea Surface Temperature	11–30°C
Ocean Depth	0–70 m below sea level

```
# Table of conditions for Maximum Osyter Growth
criteria <- tibble(
  Variable = c("Sea Surface Temperature", "Ocean Depth"),
  Suitable_Range = c("11-30°C", "0-70 m below sea level")
)

criteria |>
  kbl(
    caption = "Environmental Criteria for Oyster Suitability",
    col.names = c("Variable", "Suitable Range"),
    align = "c"
  ) |>
  kable_styling(
    latex_options = "striped",
    full_width = FALSE,
    position = "center"
  )
```

```
# reclassifying sst
rcl_sst <- matrix(c(10, 30, 1),
                  ncol = 3,
                  byrow = TRUE)
reclassified_sst <- classify(combined_rast$mean,
                             rcl = rcl_sst,
                             others = 0)

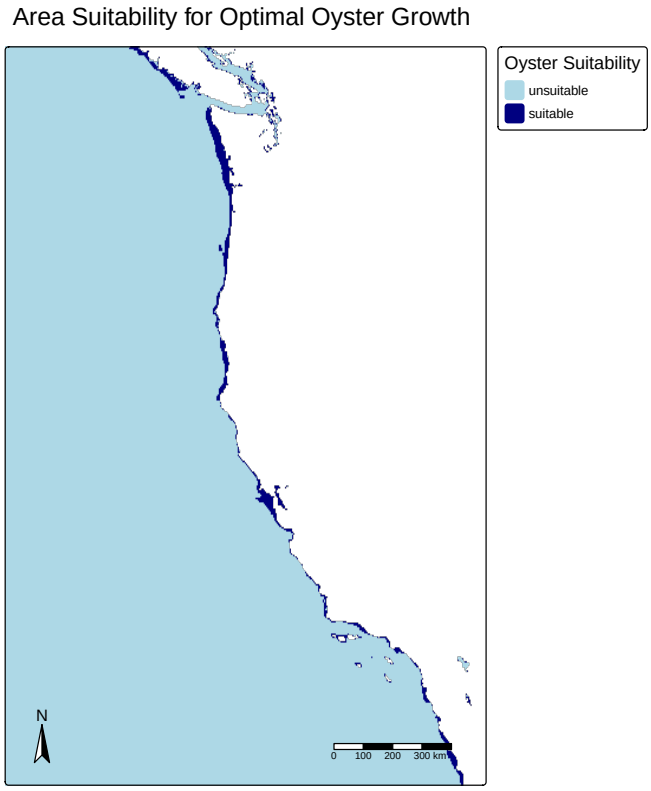

# reclassifying depth
rcl_depth <- matrix(c(-71, 0, 1),
                    ncol = 3,
                    byrow = TRUE)
reclassified_depth <- classify(combined_rast$depth,
                               rcl = rcl_depth,
                               others = 0)
```

```r
# combine reclassified values
reclassified <- c(reclassified_sst, reclassified_depth)


# function for multiplying each rows' mean and depth to determine suitability
suitability <- function(mean, depth) {
    mean * depth
}

# use lapp function to multiply values in reclassified raster using 'suitability' function
suit_conditions <- lapp(reclassified, fun = suitability)
suit_conditions[depth_cropped >= 0] <- NA

tm_shape(suit_conditions) +
    tm_raster(col.scale = tm_scale_categorical(values = c('lightblue', 'navyblue'),
                                               labels = c('unsuitable', 'suitable')),
              col.legend = tm_legend(title = 'Oyster Suitability')
    ) +
tm_title(text = 'Area Suitability for Optimal Oyster Growth') +
tm_compass(position = c('bottom', 'left')) +
tm_scalebar(position = c('bottom', 'right'), size = 0.6)
```



Reflections/Intepretations

The map above reflects the suitable areas for oyster growth in the west coast. The dark blue colors indicating that these regions along the coast meet the conditions required for optimal oyster growth. The light blue colors indicates that these regions in the ocean are not fit for oyster growth. From our map, we can see that oysters tend to have the most optimal growth along the coast rather than open ocean due to shallower depths and moderate temperatures. There is a slightly larger area more north along the west coast of the U.S.

Determine the most suitable EEZ

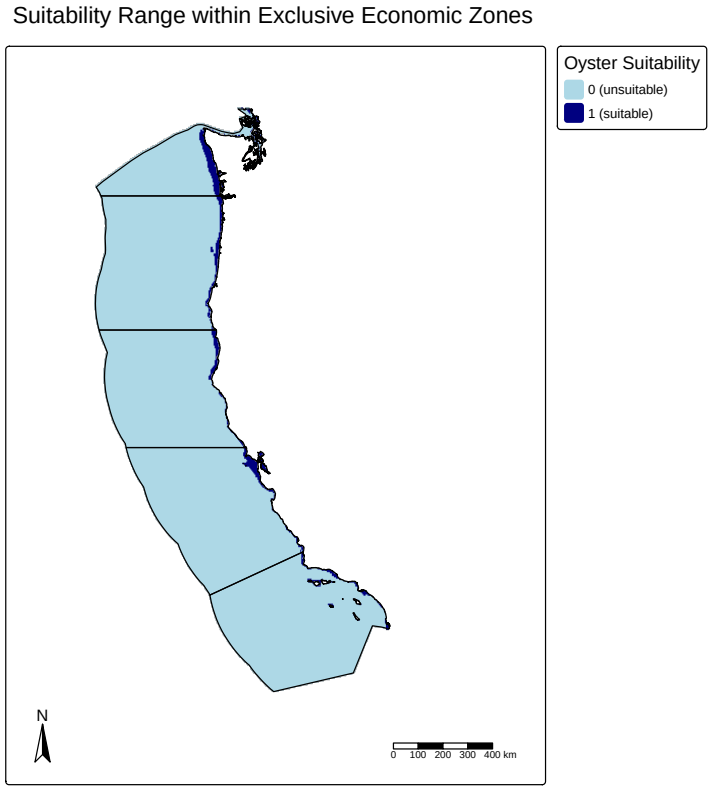
We want to determine the total suitable area within each EEZ in order to rank zones by priority. To do so, we need to find the total area of suitable locations within each EEZ.

- select suitable cells within West Coast EEZs
- find area of grid cells
- find the total suitable area within each EEZ

– hint: it might be helpful to rasterize the EEZ data

```
# change the extent of the suitable area to match the EEZ extent
west_coast <- mask(suit_conditions, eez)
```

```
# visualize the oyster suitability within each Exclusive Economic Zone
tm_shape(west_coast) +
    tm_raster(col.scale = tm_scale_categorical(values = c("lightblue", "navyblue"), labels =
    col.legend = tm_legend(title = 'Oyster Suitability')) +
tm_shape(eez) +
    tm_borders(col = 'black') +
tm_title(text = 'Suitability Range within Exclusive Economic Zones') +
tm_compass(position = c('bottom', 'left')) +
tm_scalebar(position = c('bottom', 'right'), size = 0.6) +
tm_layout(inner.margins = c(0.12, 0.05, 0.05, 0.25))
```



Reflections/Intepretations

In figure 2 we condensed the suitability range of the oyster from the previous question down to the extent of the EEZ that is separated into five different zones. The top zone appears to have the most about of area suitable for oyster growth while the bottom zone has the least suitable area for oyster growth.

```
# calculating the total area of the grid cells in west coast EEZ
total_grid_area <- cellSize(west_coast, unit = 'km')
```

```
# rasterize the eez dataframe
# The second argument (west_coast) is a template raster: the new raster copies its extent, re
eez_rast <- rasterize(eez, west_coast, field = 1:nrow(eez))


# multiplies the suitable areas (1) with the grid area
suit_area <- west_coast * total_grid_area

# calculate the total suitable area in each EEZ
zones <- zonal(suit_area, eez_rast, fun = 'sum', na.rm = TRUE)


# create a formatted table that shows the rank of EEZs by total suitable area
zones <- zones |>
    mutate(layer = c('A', 'B', 'C', 'D', 'E')) |>
    arrange(desc(lyr1)) |>
    mutate(rank = row_number())

zones |>
    kbl(
        caption = "Suitable Oyster Habitat Area by Exclusive Economic Zone",
        col.names = c("EEZ Zone", "Suitable Area (km²)", "Rank"),
        digits = 2,
        align = "c"
    ) |>
    kable_styling(
        latex_options = c("striped", "hold_position"),
        full_width = FALSE,
        position = "center"
    )
```

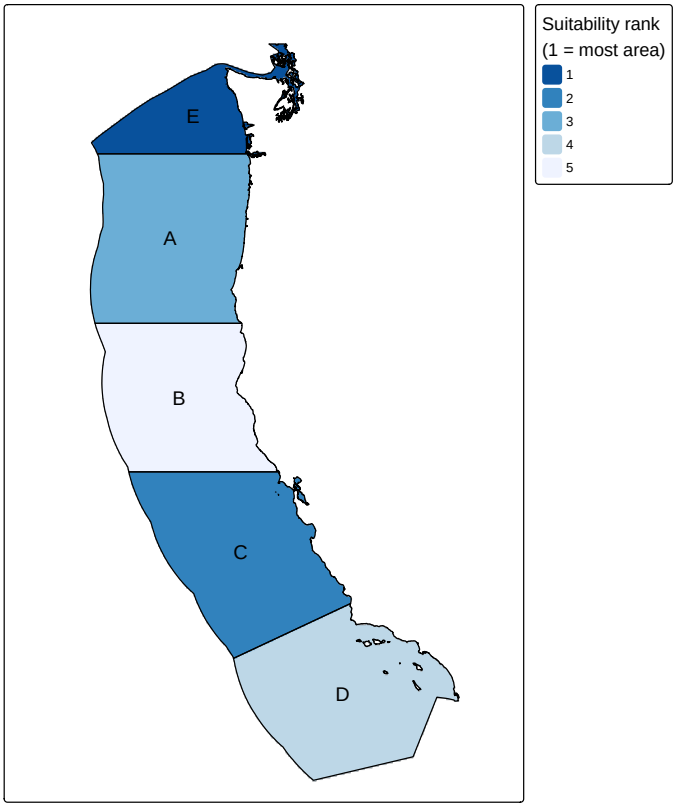
Table 2: Suitable Oyster Habitat Area by Exclusive Economic Zone

EEZ Zone	Suitable Area (km^2)	Rank
E	6403.09	1
C	4712.85	2
A	4067.20	3
D	3347.62	4
B	2849.30	5

```
# ranking each zone within the EEZ by best area for oyster growth
rcl_rank <- matrix(c(5, 1,
                     3, 2,
                     1, 3,
                     4, 4,
                     2, 5),
                   ncol = 2, byrow = TRUE)

rank_rast <- classify(eez_rast, rcl = rcl_rank)
eez$zone <- c("A", "B", "C", "D", "E")

tm_shape(rank_rast) +
  tm_raster(col.scale = tm_scale_categorical(values = "-blues"),
            col.legend = tm_legend(title = "Suitability rank\n(1 = most area)")) +
tm_shape(eez) +
  tm_borders(col = 'black') +
  tm_text('zone')
```




```
tm_title(text = 'EEZ Oyster Suitability Ranking') +
tm_compass(position = c('bottom', 'left')) +
tm_scalebar(position = c('bottom', 'right'), size = 0.6) +
tm_layout(inner.margins = c(0.12, 0.05, 0.05, 0.25))
```

Reflections/Intepretations

This graph ranks each of the zones by how much more suitable area the zone contains for oysters. The darker blue represents higher priority for marine aquaculture while the lighter shades indicates lower priority. Our initial observation for the lowest area coverage was inaccurate as from the graph we can see that the middle zone had the lowest with a total area of 2849.3 km. However our observation for the top zone was accurate as it had the highest suitability area coverage with a total area of 6403.088 km.

Data Citation

Data	Citation	Link
Sea Surface Temperature 2008-2012	NOAA Coral Reef Watch. (2018, updated daily). *NOAA Coral Reef Watch Version 3.1 Daily Global 5km Satellite Sea Surface Temperature Anomaly Product*. College Park, Maryland, USA: NOAA Coral Reef Watch.	NOAA Coral Reef Watch
Bathymetry	GEBCO Bathymetric Compilation Group 2026. (2026). *The GEBCO_2026 Grid – a continuous terrain model for oceans and land at 15 arc-second intervals*.	GEBCO Bathymetry
Exclusive Economic Zones	Flanders Marine Institute. (2023). *Maritime Boundaries Geodatabase: Maritime Boundaries and Exclusive Economic Zones (200NM), version 12*.	Marine Regions